Enterprise Customer Onboarding Agent — Solution Design Document

April 2026

1. Architecture Overview

The solution implements an **AI-powered automation agent** that orchestrates enterprise customer onboarding from deal closure through SaaS provisioning. Built with **Pydantic AI** for agentic tool calling and **FastAPI** for the REST interface, the agent uses native LLM tool calling to autonomously reason through 28 registered tools. Each tool is mirrored as a **FastMCP** server definition for MCP-style extraction. A second **CS Assistant Agent** shares all tools but returns free-form text for conversational interaction via a **Streamlit UI** with four pages (Dashboard, Portfolio Overview, Run Onboarding, Chat). Data flows sequentially across systems via **cross-system chaining**: Opportunity.ContractId → SF Contract.Id → CLM.salesforce_contract_id → Invoice.clmContractRef, with fallback to account-based lookups when chain links are missing. The latest prototype also closes the CS action loop: suggested actions can simulate upstream issue resolution in mocked SaaS systems and then re-run onboarding so a previously blocked or escalated account can actually progress into provisioning.

1.1 System Integrations

System	Objects & Key Fields
Salesforce CRM	`Account` (Id, Name, IsDeleted, OwnerId), `Opportunity` (StageName, Amount, **ContractId**), `Contract` (**Id**, Status, StartDate), `User` (IsActive, Email). [API Ref]
CLM System	contract_id, status, **salesforce_contract_id** (chains from SF Contract.Id), signatories, key_terms. Mock simulating DocuSign CLM.
NetSuite ERP	`Invoice` (status, dueDate, amountRemaining, total, **clmContractRef** chains from CLM contract_id). [API Ref]
Provisioning	Tenant creation + 14-task onboarding checklist with dependencies, owners, and due dates.
Currency (Frankfurter)	Live and historical exchange rates via ECB data. Converts invoice amounts across currencies (e.g., CAD↔USD) using the invoice date rate. [API Ref]
Sentiment (ML)	DistilBERT transformer model (`distilbert-base-uncased-finetuned-sst-2-english`) via PyTorch + HuggingFace for customer interaction scoring. Keyword fallback when ML stack unavailable.

Field Selection Rationale: `Account.IsDeleted` validates account exists; `Opportunity.StageName="Closed Won"` confirms deal closure; `Invoice.status` identifies payment blockers; `CLM.status` in {EXECUTED, SIGNED} required for provisioning. Currency conversion uses historical rates from the invoice date for financial accuracy. Minimizes API calls while capturing decision-critical data.

1.2 Data Flow

(1) Webhook with `account_id` + `correlation_id` **(2)** Fetch account, then user/opportunity in parallel; chain sequentially: Opp.ContractId → SF Contract → CLM (via salesforce_contract_id) → Invoice (via clmContractRef) **(3)** `validate_business_rules` + `check_financial_alignment` **(4)** PROCEED → provision + notify; BLOCK/ESCALATE → notify, remediate, and optionally re-run **(5)** PROCEED only: mandatory `check_onboarding_progress` + `identify_onboarding_risks` + `get_customer_sentiment` for post-provisioning risk and sentiment assessment.

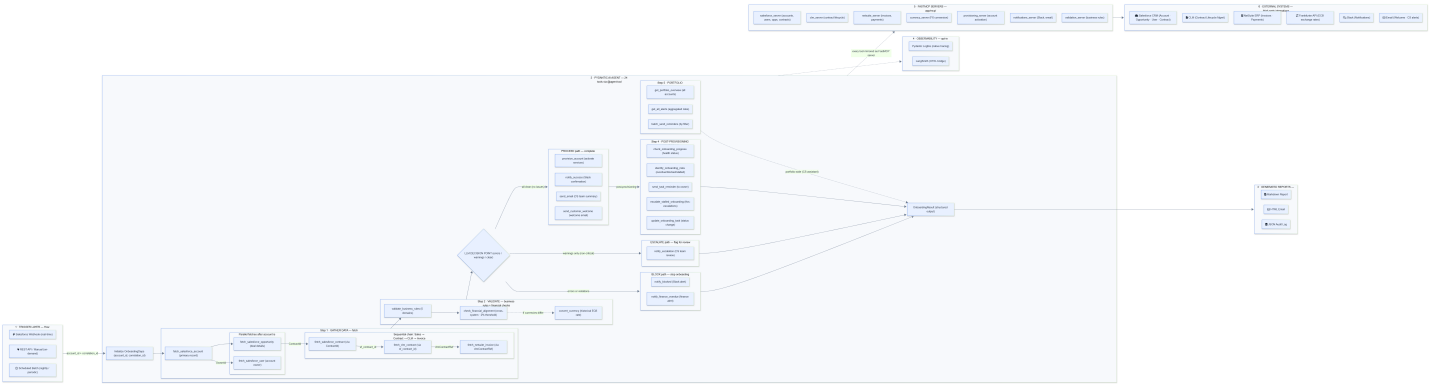

Figure 1: Architecture: triggers, Pydantic AI agent orchestration with 28 tools, cross-system data chaining, FastMCP servers, and external system integrations.

2. AI Agent Application

The agent leverages **OpenAI GPT-4o-mini** (configurable via `OPENAI_MODEL`) with **Pydantic AI** for native tool calling. The LLM autonomously decides which of the 28 registered tools to call, in what order, and how many times. Falls back to **Ollama** local models when no API key is available. A second **CS Assistant Agent** shares all tools with free-form text output for conversational monitoring via the Streamlit UI.

2.1 LLM-Powered Intelligence

- **Risk Analysis**: Evaluates API errors, violations, warnings → risk levels (low/medium/high/critical) with business impact and resolution time estimates.
- **Summary Generation**: Human-readable reports for CS teams, translating technical states to actionable insights.
- **Action Recommendations**: Prioritized steps with owner assignment (CS, Finance, IT/DevOps, Legal, Sales Ops).
- **Error Interpretation**: Converts technical codes (e.g., `INVALID_SESSION_ID`) to plain English with resolution steps.

2.2 Autonomous Actions

(1) Auto-provisions the SaaS tenant with tier-appropriate configuration (Enterprise/Growth/Starter).
(2) Creates a 14-task onboarding checklist with dependencies, ownership (system/cs_team/customer), and due dates.
(3) Sends Slack notifications to `#cs-onboarding-alerts` (blocked) or `#cs-onboarding` (success/escalate).
(4) Sends customer-facing welcome emails with tenant ID and login URL.
(5) Generates HTML/Markdown/JSON reports for auditing and traceability.
(6) Post-provisioning: monitors progress, detects risks, sends reminders, escalates stalled onboardings.
(7) Portfolio management: aggregated alerts across accounts, suggested actions with approve/dismiss, batch reminders.
(8) Remediation loop: the CS assistant can simulate source-system fixes (deal closed, contract executed, invoice paid, missing record created) and safely re-run onboarding from BLOCK/ESCALATE into PROCEED.
(9) Sentiment analysis: ML-powered scoring of customer interactions via DistilBERT (PyTorch + HuggingFace) with trend detection; runs as a mandatory step (step 5) after every provisioning alongside risk assessment; negative sentiment shifts health to at_risk and triggers proactive outreach suggestions.
(10) Product assistance: looks up tier configuration (features, limits, storage), SLA details, contractual terms, and implementation prerequisites from CLM and provisioning data.

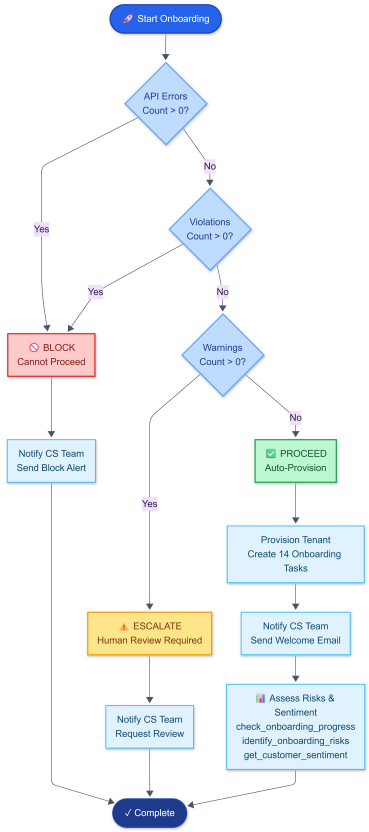

Figure 2: Decision: API Errors/Violations → BLOCK, Warnings → ESCALATE, Clear → PROCEED.

2.3 Onboarding Task Management

The provisioning system creates a 14-task onboarding checklist:

Task Categories:

- **Automated (4)**: Tenant creation, API credentials, welcome email, training materials
- **CS Team (5)**: Schedule/conduct kickoff, SSO config, custom reports, 30-day check-in
- **Customer (4)**: Verify login, complete tour, invite team, create first campaign
- **Milestone (1)**: Onboarding complete marker

API Endpoints:

- `GET /tasks/{id}` – All tasks with status
- `GET /tasks/{id}/pending?owner=cs_team` – Filtered pending
- `GET /tasks/{id}/overdue` – Overdue task alerts
- `PUT /tasks/{id}/{task_id}` – Update task status
- `GET /tasks/{id}/next-actions` – Ready-to-execute tasks

Tasks include dependencies (e.g., "Complete Tour" depends on "Verify Login"), due dates calculated relative to provisioning, and ownership assignment enabling filtered views per team.

2.4 Sentiment Analysis as a Proactive Risk Signal

Beyond task and billing state, the CS assistant also tracks **customer sentiment** from inbound and outbound interactions such as email, chat, support tickets, and calls. Sentiment scoring uses a **DistilBERT transformer model** (`distilbert-base-uncased-finetuned-sst-2-english`) via **PyTorch + HuggingFace Transformers** for real ML inference, with graceful fallback to keyword-based scoring when the ML stack is not installed. The model is lazy-loaded on first use to avoid impacting startup time. The prototype exposes two sentiment tools: `log_customer_interaction` records structured interactions for an account, and `get_customer_sentiment` returns a score, label (positive/neutral/negative), trend (improving/stable/declining), and the model used. This allows the agent to identify onboarding risk *before* a task is formally overdue or blocked.

Sentiment Signal	Agent Behavior
Negative sentiment + declining trend	Recommend proactive outreach, schedule a check-in call, raise account priority in portfolio alerts
Negative sentiment + blocked onboarding tasks	Escalate urgency because operational blockers and customer frustration are compounding
Improving sentiment after outreach	Reduce risk posture and keep onboarding in the monitored PROCEED path

2.5 Business Rule Validation (Invariants)

The agent enforces a two-tier validation system across five domains. **Violations** are blocking (cause BLOCK decision); **Warnings** are non-blocking (cause ESCALATE for human review).

Domain	Blocking Violations	Non-blocking Warnings
Account	Missing Id/Name, IsDeleted=true	Missing BillingCountry, Industry, OwnerId
Opportunity	Stage $\neq$ "Closed Won", Missing Id/AccountId	Missing Amount, CloseDate, ContractId
Contract (CLM)	Status $\notin$ {EXECUTED, SIGNED}, Missing contract_id	No effective/expiry date, pending signatories
Invoice	Voided, Cancelled, Missing invoice_id	Overdue, Pending, <50% paid, missing due_date
User	Inactive, Missing Id/Email/ProfileId	Missing Title, Department, ManagerId

3. Orchestration & Event-Driven Flows

Trigger	Source	Endpoint
Webhook	Salesforce Flow	`POST /webhook/onboarding`
Manual	CS Team	`POST /demo/run/{account_id}`
Batch	Scheduler	`POST /demo/run/{account_id}` (per account)

Error Simulation: `/demo/enable-random-errors` injects failures (401 auth, 400 validation, 429 rate limit, 500 server) at configurable rates for chaos testing without modifying business logic.

Report Generation: Each run produces three output files:

- `run_report_*.md` – Full Markdown report with all details
- `email_*.html` – Professional HTML email (blocked, escalation, success, or welcome)
- `audit_*.json` – Machine-readable audit log

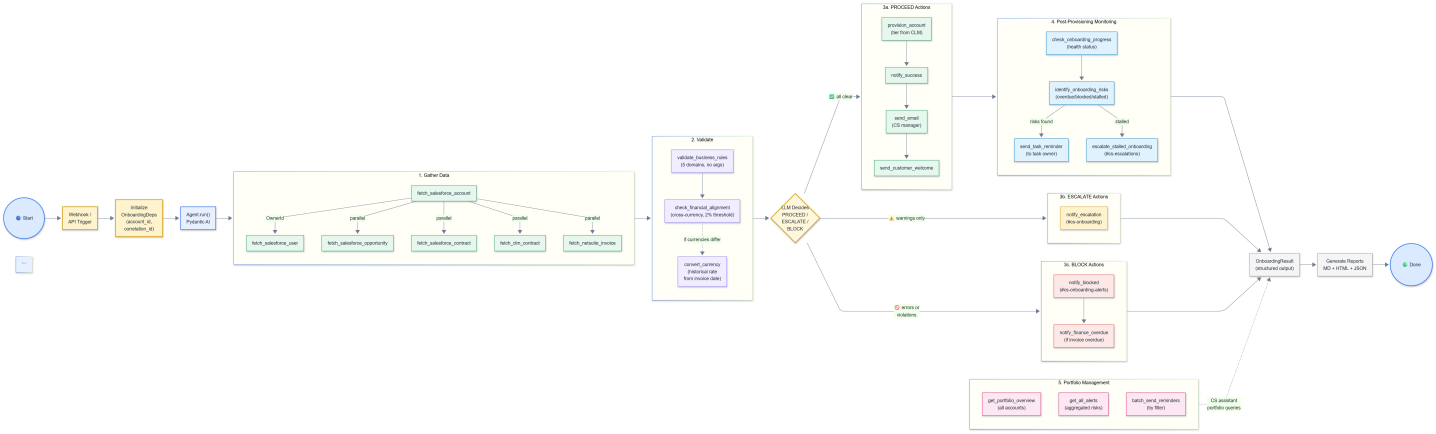

Figure 3: Agent workflow: full lifecycle from webhook ingestion through tool calling, validation, decision routing, post-provisioning monitoring, and portfolio management.

3.1 API Error Classification & Handling

Each integration returns structured error payloads with `error_type`, `error_code`, `http_status`, `message`, `resolution`, and `owner`:

Error Type	HTTP	Resolution
Authentication	401	Re-authenticate, refresh token (IT/DevOps)
Authorization	403	Check profile/permission sets (System Admin)
Validation	400	Fix field values in source (Data Admin)
Rate Limit	429	Implement backoff, increase quota (Integration Admin)
Server Error	500	Check status page, retry later (Support Team)

3.2 Error Handling Features

- **In-place Error Simulator**: `ERROR_SIMULATOR` object modified in-place when enabled; all modules reference the same instance
- **Comprehensive Catching**: Each integration catches specific error types AND generic `APIError` as fallback
- **Error-Aware Decisions**: `make_decision` checks `api_errors` first — system failures block onboarding
- **Full Traceability**: API errors added to violations and appear in reports with context

4. Trade-offs, Assumptions & Considerations

4.1 Key Trade-offs

Decision	Trade-off
API Errors → BLOCK	Integrity over speed
Sequential Data Fetching	Clearer failure tracing but has higher latency
Rule-based Fallback	Less smart but reliable
14 Fixed Tasks	Predictable but rigid
LLM Advisory Only	Auditable but less adaptive

4.2 Assumptions

- Salesforce is source of truth for accounts
- CLM status reflects actual signatures
- Invoice payment status is current
- Cross-currency comparison uses historical ECB rates from invoice date with 2% threshold
- One invoice per account per onboarding run
- One Opportunity per Account (no multi-deal)
- Contract has a single version, no amendments
- CS monitors Slack channels for alerts
- Single onboarding per account at a time
- Customer email available in CLM signatories
- Account ID consistent across all systems

4.3 Limitations

- **Mock integrations**: Salesforce, NetSuite, CLM use simulated data; not connected to real APIs
- **In-memory state**: Provisioning and task data lost on restart; no persistent database
- **Single instance**: No horizontal scaling or load balancing
- **Limited approval UI**: Suggested Actions panel provides approve/dismiss and real backend execution; full Slack interactive approval with identity/policy controls not yet implemented
- **No webhook auth**: Inbound webhooks not verified via HMAC or shared secret

4.4 Testing & Demonstration

The system includes 10 pre-configured demo scenarios validating all decision paths. Three PROCEED scenarios include post-provisioning simulation to demonstrate proactive risk detection, and five BLOCK/ESCALATE scenarios now support remediation plus onboarding re-run to demonstrate closed-loop CS intervention:

Account ID	Scenario	Description	Decision
ACME-001	Happy Path	All checks pass, contract executed, invoice paid	PROCEED
BETA-002	Opportunity Blocked	Opportunity in "Negotiation" stage	BLOCK
GAMMA-003	Overdue Invoice	Invoice 25+ days overdue, needs finance review	ESCALATE
DELETED-004	Deleted Account	Account marked `IsDeleted=true`	BLOCK
MISSING-999	Account Not Found	Account does not exist in any system	BLOCK
FOREX-005	FX Invoice Mismatch	CAD $145k invoice vs USD $100k opportunity; historical rate gap >2%	ESCALATE
PARTIAL-006	Partial Payment Gap	$190k paid of $200k invoice (5% underpayment exceeds 2% threshold)	ESCALATE
STARTER-007	Customer Not Logged In	All checks pass; post-provisioning: customer hasn't logged in after 5 days	PROCEED
GROWTH-008	Stalled Onboarding	All checks pass; post-provisioning: <30% complete after 10 days, kickoff not done	PROCEED
ENTERPRISE-009	SSO & Blocked Tasks	All checks pass; post-provisioning: SSO blocked, multiple tasks overdue after 8 days	PROCEED

Risk and action scenarios after provisioning:

- `STARTER-007`: detects customer inactivity (no login after 5 days) and recommends or executes a login reminder.
- `GROWTH-008`: detects onboarding stalling (<30% complete after 10 days) and escalates to CS management.
- `ENTERPRISE-009`: detects blocked SSO work plus overdue customer tasks and recommends SSO follow-up, task reminders, or escalation.
- `Sentiment scenario`: repeated negative customer interactions produce a declining sentiment trend; the CS assistant recommends a proactive call and the suggested-action flow can execute that outreach.
- `BETA-002`, `DELETED-004`, `MISSING-999`, `FOREX-005`, `PARTIAL-006`: support simulated remediation of source-system blockers/warnings followed by a fresh onboarding run into provisioning.

4.5 Observability & Tracing

The agent supports **dual opt-in tracing** — both can run simultaneously:

- **Pydantic Logfire**: Native agent tracing — reasoning chains, tool calls with inputs/outputs, structured output validation.
- **LangSmith**: Agent execution runs, tool invocations, LLM completions via OpenTelemetry bridge (`OtelSpanProcessor`). Logfire owns the TracerProvider; LangSmith attaches as a span processor.
- **Structured JSON Logging**: All events logged with `correlation_id`, timestamps, and context.

4.6 Security Patterns

All integrations are mocked but demonstrate production security patterns:

- **OAuth Simulation**: Token expiry and refresh flows with distinct error types for expired vs invalid tokens
- **Permission Checking**: Role-based access control verified before every API call
- **Credential Validation**: Structured error hierarchy (`AuthenticationError`, `AuthorizationError`, `ValidationError`, `RateLimitError`)
- **Audit Logging**: Correlation IDs propagated end-to-end for traceability across all systems
- **Error Masking**: No sensitive data (tokens, secrets) exposed in API responses or reports

5. Multi-Agent Architecture via Model Context Protocol (MCP) Server Connections

Note: All 28 agent tools are mirrored as FastMCP server definitions in `app/mcp/` (8 servers: Salesforce, CLM, NetSuite, Currency, Provisioning, Notifications, Validation, Sentiment). These can be extracted to standalone MCP services for multi-agent collaboration.

The architecture supports multi-agent collaboration through the **Model Context Protocol (MCP)**, enabling specialized agents to interact through standardized, governed tool interfaces rather than direct API access. Each integration is wrapped as a FastMCP server, cleanly separating decision logic from execution concerns.

Agent	Responsibility & MCP Tools
Coordinator	Orchestrates end-to-end workflow (`salesforce.*`, `provision.*`)
Contract Agent	Monitors contract execution and signatures (`clm.get_contract`)
Finance Agent	Evaluates billing, payments, and cross-currency alignment (`netsuite.*`, `currency.*`)
Task Monitor	Detects overdue onboarding tasks (`tasks.get_overdue`)
Sentiment Agent	Scores customer interactions via ML inference (`sentiment.*`)

Benefits: Independent versioning per integration • Clear domain ownership • Centralized access control and audit logging • Scalable agent complexity • Cross-agent context sharing.

Concurrency & API Execution Strategy: MCP servers act as controlled execution layers for external APIs, enabling concurrent and batch-aware request handling. Batch or composite API calls are leveraged where supported (e.g., Salesforce composite reads), with fallback to bounded parallel single-entity requests. Concurrency limits, rate limiting, and circuit breakers are enforced at the MCP layer to prevent downstream overload and cascading failures.

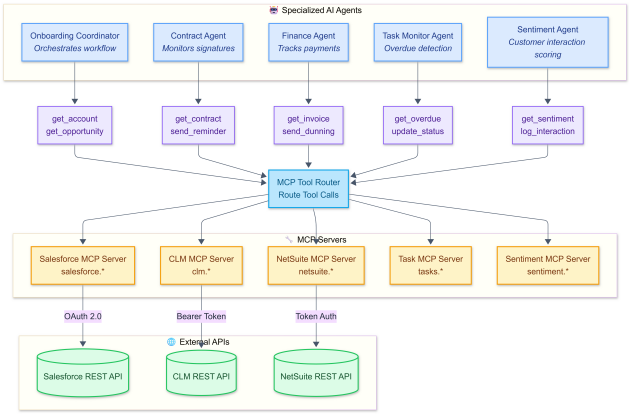

Figure 4: MCP: Specialized agents collaborate through shared execution servers wrapping external APIs.

6. Production Considerations

For additional production enhancements, see the `production_roadmap` directory in the repository.

Cloud Deployment & CI/CD

Frontend: Streamlit UI with four pages (Dashboard, Portfolio, Run Onboarding, Chat). Production: migrate to React/ASP.NET with auth and WebSocket updates.
Infrastructure: ECS/Fargate with horizontal scaling. **AWS Bedrock** for multi-model LLM inference (Claude, Llama, Titan) with IAM auth. Docker + Kubernetes for consistent environments, auto-scaling, and zero-downtime deployments.
CI/CD: GitHub Actions / CodePipeline — unit tests for invariants, risk analysis, and decision routing on every PR. Reproducible, auditable deployments with safe rollback.

Observability: Dual tracing (Pydantic Logfire + LangSmith via OpenTelemetry). Prometheus metrics, DataDog/Jaeger distributed tracing, PagerDuty alerts.

Agentic Execution & API Governance

Deterministic Boundaries: LLM reasons through tools via `@agent.tool`, but `validate_business_rules` and `check_financial_alignment` define all violations/warnings. LLM routes (PROCEED/ESCALATE/BLOCK) based on these outputs.
Data Chaining: Fetch account → user/opportunity in parallel → sequential chain: Opp.ContractId → SF Contract → CLM → Invoice. Fallback to account-based lookups. Production: batch APIs, concurrent fetches, bounded backoff.
Idempotency & Retry: Runs keyed by `correlation_id` for safe replays. Retryable errors (5xx, 429) → exponential backoff; non-retryable → immediate halt. Circuit breakers prevent cascading failures.

Governance: PII-masked data sent to LLMs. Per-source rate limits. Agent version, rule set, and prompt version recorded per run. All 28 tools mirrored as 8 FastMCP servers, extractable to standalone services with A2A protocol.